The Tsetlin machine goes deep

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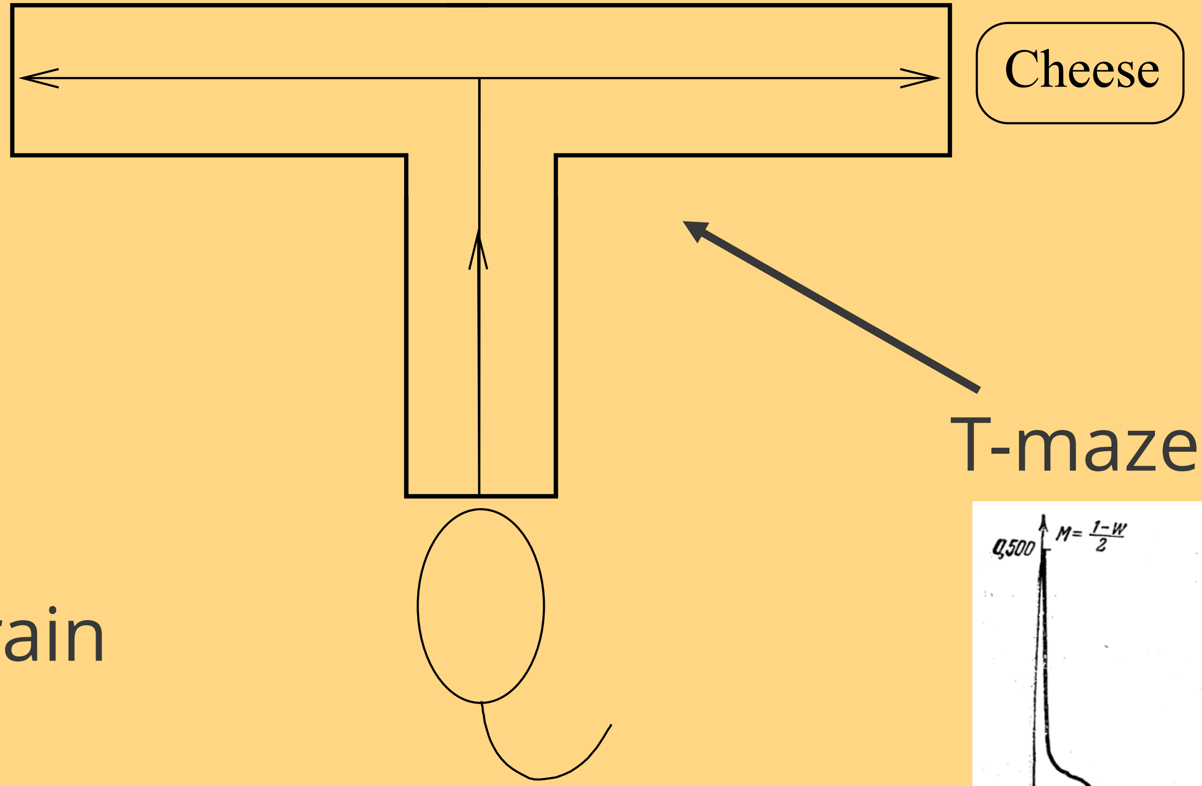
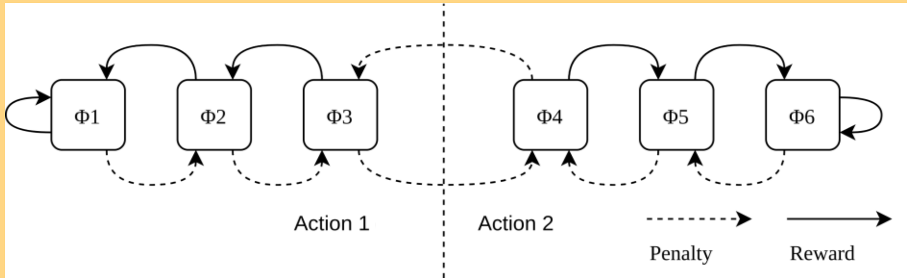
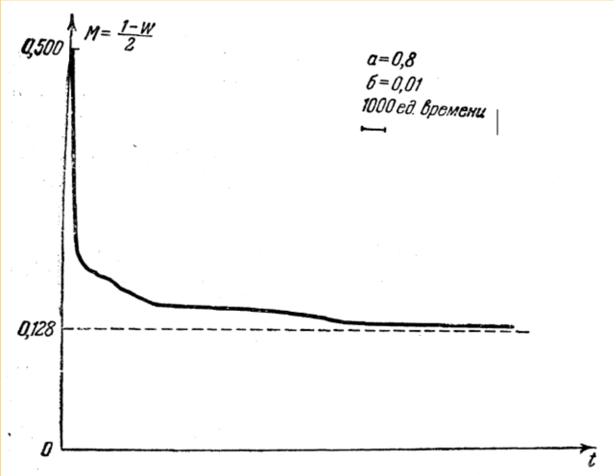
The Tsetlin automaton:
A mathematical rat brain



T-maze puzzle

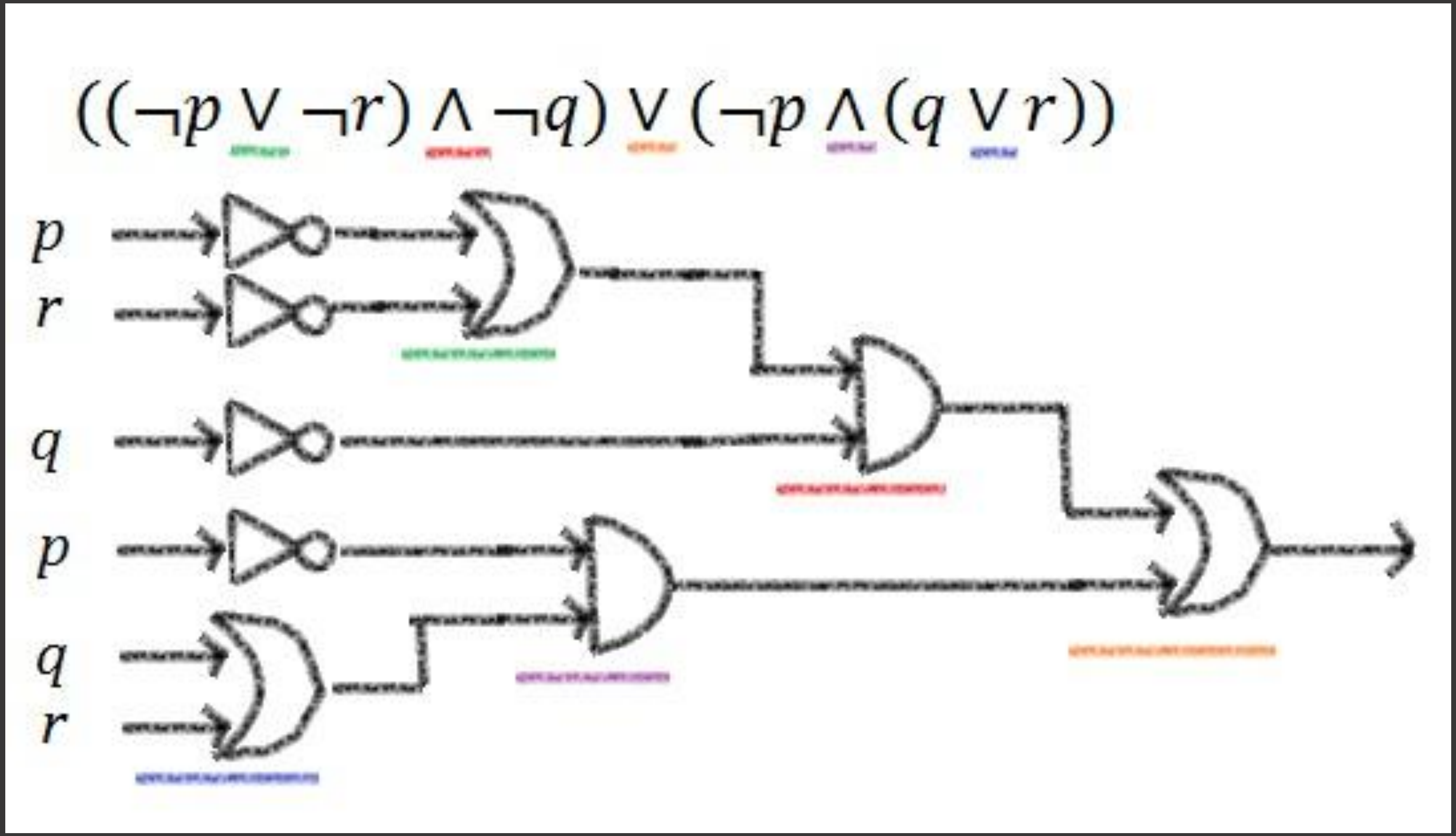
Mathematical rat brain





Tsetlin, Michael L. (1961). "On behaviour of finite automata in random medium". Avtomat. i Telemekh. 22 (10).

Propositional logic



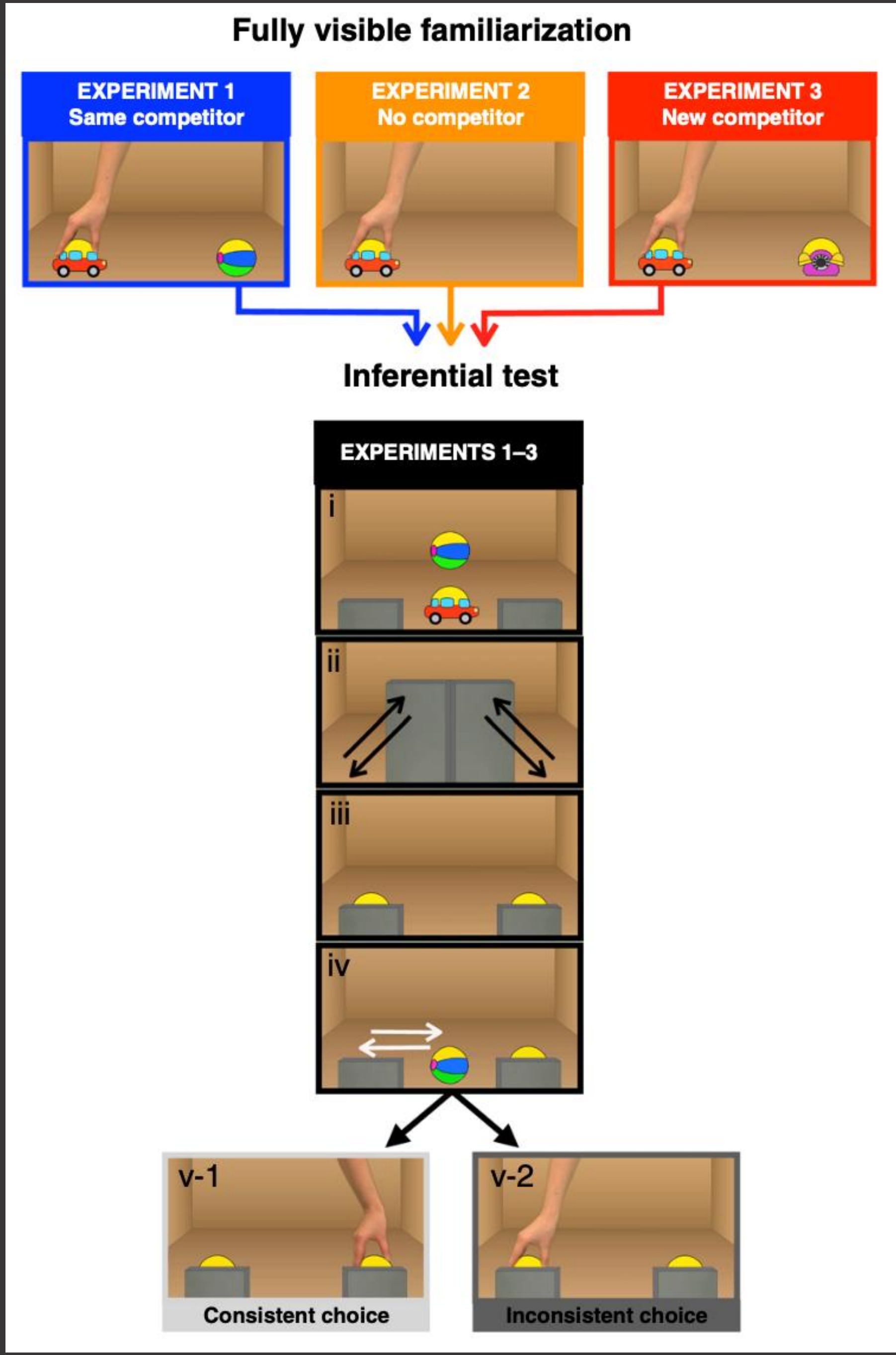
http://mathrescue.blogspot.no/2013/04/

Infants recruit logic to learn about the social world

Reasoning by elimination

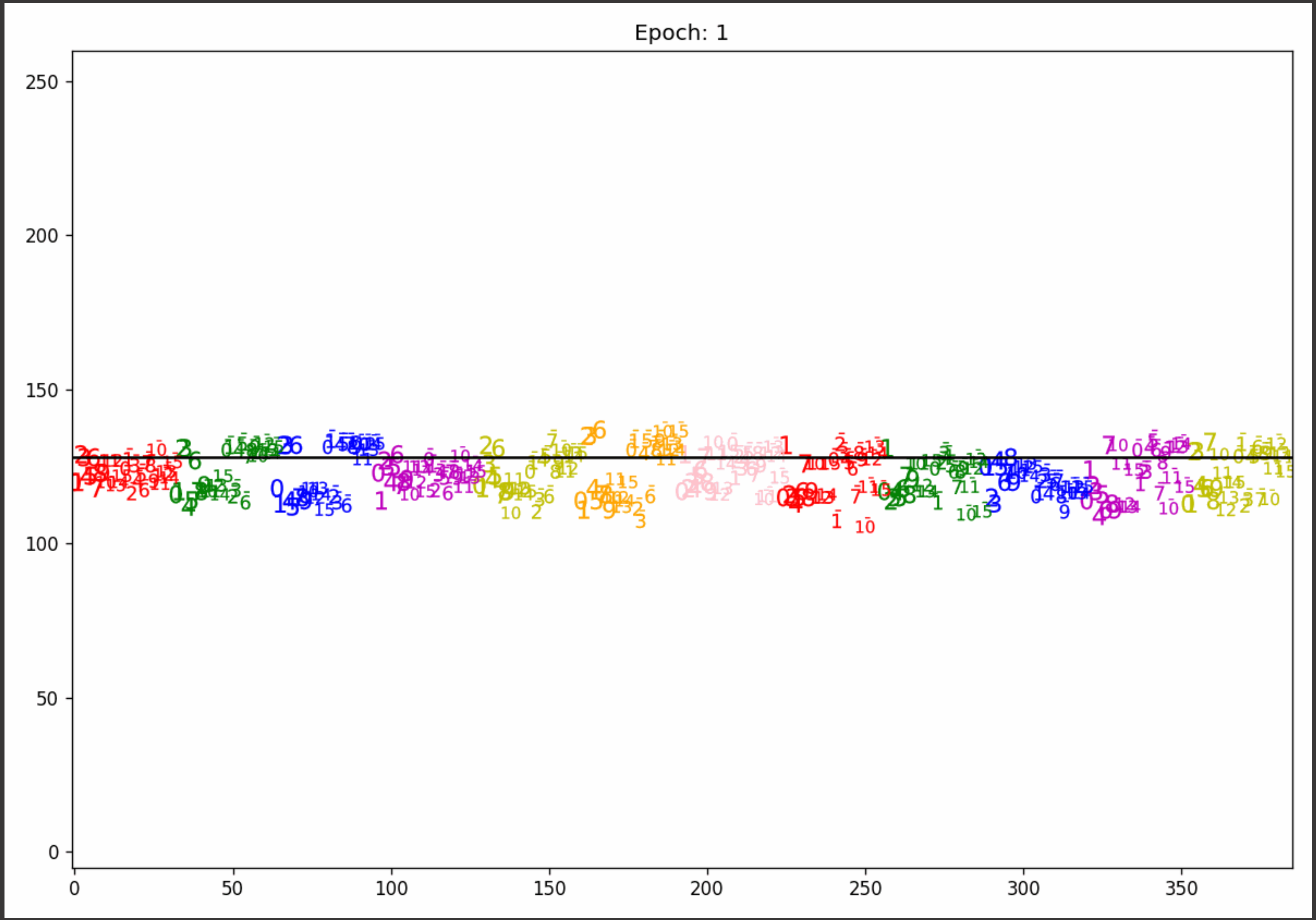
A **or** B
not A

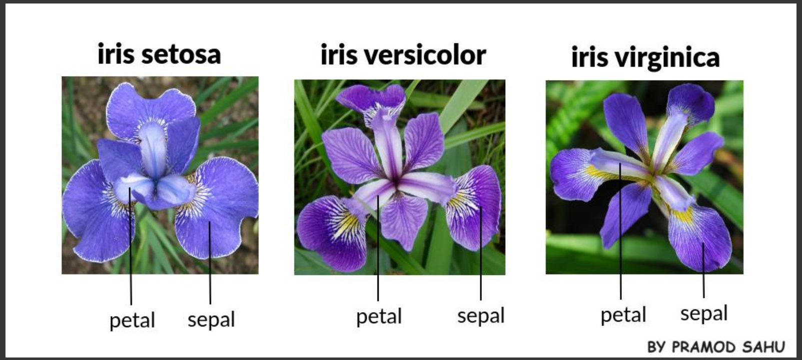
B



N. Cesana-Arlotti, A. M. Kovacs, and E. Teglas. Infants recruit logic to learn about the social world. Nature Communications, 11(1):5999, Nov 2020.

Coordinated learning





By Runar Helin, 2024

Noisy multi-valued XOR problem

$$< \text{Even} > \wedge < \text{Even} > \rightarrow 0$$
$$< \text{Even} > \wedge < \text{Odd} > \rightarrow 1$$
$$< \text{Odd} > \wedge < \text{Even} > \rightarrow 1$$
$$< \text{Odd} > \wedge < \text{Odd} > \rightarrow 0$$

Multi-valued XOR Problem

= deep solution

Flat brute-force solution

$$
\begin{array}{lll}
C_1 = 1 \wedge 2 & C_6 = 2 \wedge 1 & C_{46} = 10 \wedge 1 \\
C_2 = 1 \wedge 4 & C_7 = 2 \wedge 3 & C_{47} = 10 \wedge 3 \\
C_3 = 1 \wedge 6 & C_8 = 2 \wedge 5 \cdots & C_{48} = 10 \wedge 5 \\
C_4 = 1 \wedge 8 & C_9 = 2 \wedge 7 & C_{49} = 10 \wedge 7 \\
C_5 = 1 \wedge 10 & C_{10} = 2 \wedge 9 & C_{50} = 10 \wedge 9
\end{array}
$$

50 Flat Conjunctive Clauses

Deep solution – concept building

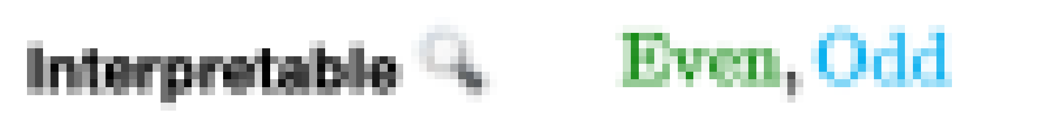
Even $C_1^0 = 2 \vee 4 \vee 6 \vee 8 \vee 10$

Odd $C_2^0 = 1 \vee 3 \vee 5 \vee 7 \vee 9$

Deep solution – composition

Deep Clause 🔥 $C_1 = C_1^0 \wedge C_2^0$

Result: Interpretable deep learning



Unfortunately, the Tsetlin machine can only create AND-clauses... Right?

Or...

De Morgan's laws

$$\neg(P \land Q) \vdash (\neg P \lor \neg Q), \text{and}$$
$$(\neg P \lor \neg Q) \vdash \neg(P \land Q).$$

Negation of conjunctions

https://en.wikipedia.org/wiki/De_Morgan%27s_laws

The Tsetlin machine can do ***double negation****…*

Problem solved!

$$\text{Even} = \neg\text{Odd}$$
$$\text{Odd} = \neg\text{Even}$$

Layer-zero clause components

Clause Components Layer Zero

Even $C_1^0 = \neg 1 \land \neg 3 \land \neg 5 \land \neg 7 \land \neg 9$

Odd $C_2^0 = \neg 2 \land \neg 4 \land \neg 6 \land \neg 8 \land \neg 10$

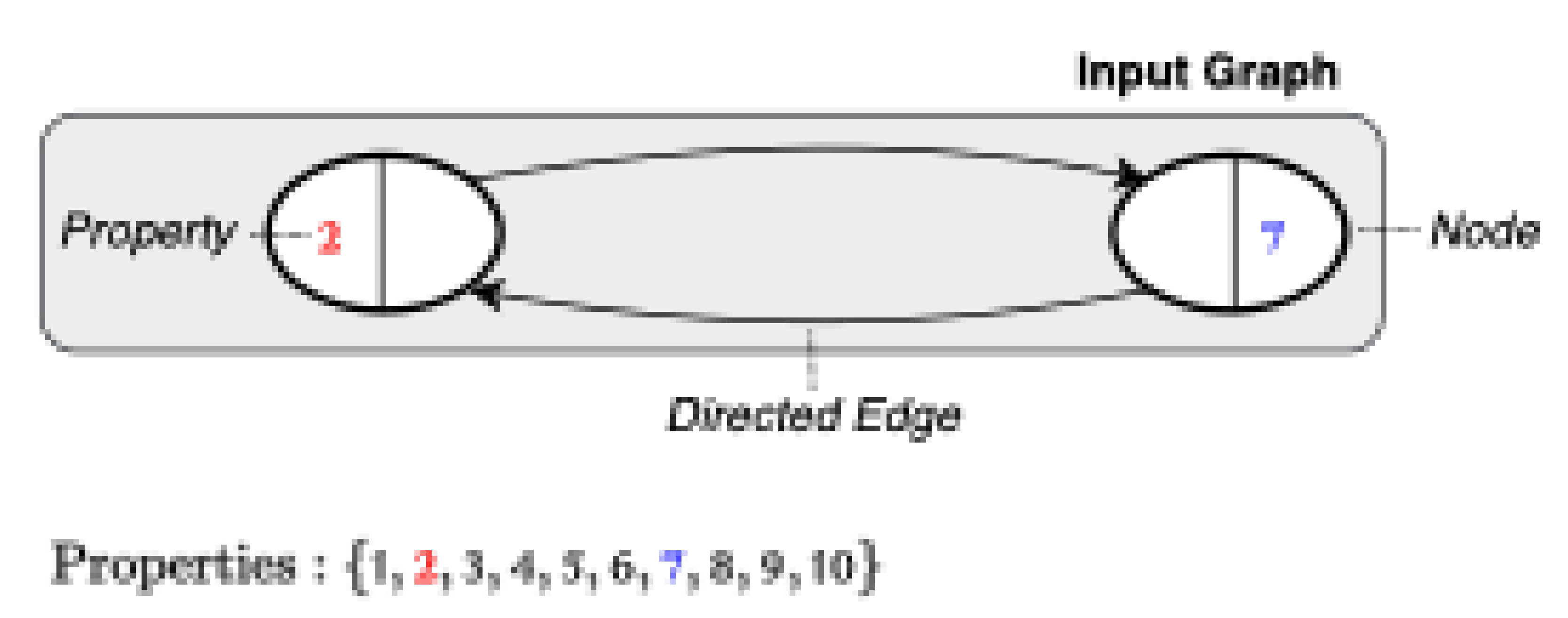
Reasoning by Elimination

Finally: Two-layered clauses

$$C_1 = C_1^0 \wedge C_3^0 \qquad \text{Even, Odd}$$

How to represent relations between lower-layer concepts?

Answer: Graph representation...



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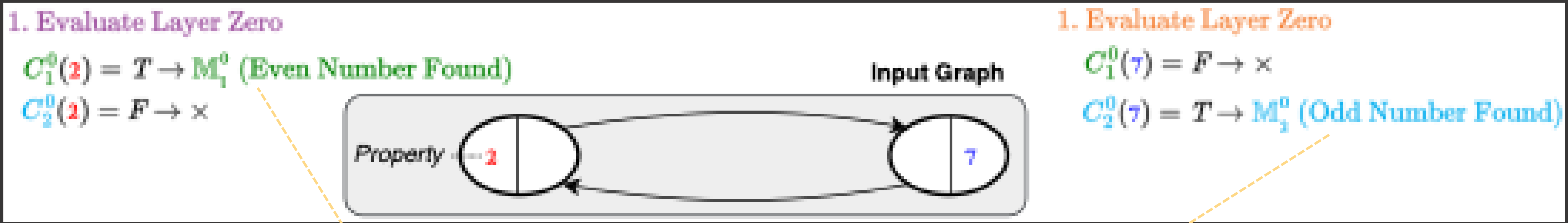
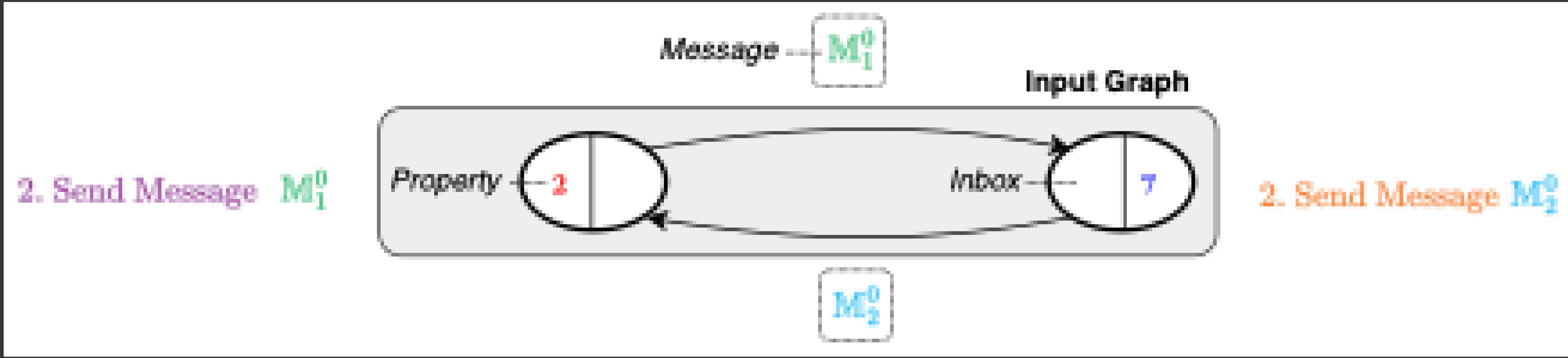
Reasoning

Step 1: Evaluate layer zero



Clauses that match a node create a message

Step 2: Message passing



Two-layered clauses

Clause Components Layer Zero

Even $C_1^0 = \neg 1 \land \neg 3 \land \neg 5 \land \neg 7 \land \neg 9$

Odd $C_2^0 = \neg 2 \land \neg 4 \land \neg 6 \land \neg 8 \land \neg 10$

Reasoning by Elimination

Clause Components Layer One

Odd Number in Neighbor $C_1^1 = \mathrm{M}_2^0$

Even Number in Neighbor $C_2^1 = \mathrm{M}_1^0$

Two-layered Clauses

$C_1 = C_1^0 \land C_1^1$ Even, Odd

$= \neg 1 \land \neg 3 \land \neg 5 \land \neg 7 \land \neg 9 \land \mathrm{M}_2^0$

$C_2 = C_2^0 \land C_2^1$ Odd, Even

$= \neg 2 \land \neg 4 \land \neg 6 \land \neg 8 \land \neg 10 \land \mathrm{M}_1^0$

Step 3: Evaluate layer one

3. Evaluate Layer One

$$C_1(2, M_2^0) = C_1^0(2) \land C_1^1(M_2^0) = T \land T = T$$
$$C_2(2, M_2^0) = C_2^0(2) \land C_2^1(M_1^0) = F \land F = F$$

3. Evaluate Layer One

$$C_1(7, M_1^0) = C_1^0(7) \land C_1^1(M_2^0) = F \land F = F$$
$$C_2(7, M_1^0) = C_2^0(7) \land C_2^1(M_1^0) = T \land T = T$$

$C_1 = C_1^0 \land C_1^1$ Even, Odd

Local Incoming

$C_2 = C_2^0 \land C_2^1$ Odd, Even

Local Incoming

Node = viewpoint on information

Step 4: Output voting & clause feedback

Very simple…
Standard clause voting and feedback (cf. convolution)

Vector symbolic representation (one-hot, thermometers, or hypervectors)

Vector symbolic node properties

Two randomly set bits define the symbol A in hyperspace

A – [0 0 1 0 0 0 0 1]

Symbol

Rachkovskij DA (2001) Representation and processing of structures with binary sparse distributed codes. IEEE Trans Knowl Data Eng 13(2):261–276.

Bundling node properties

$$A := [0\ 0\ 1\ 0\ 0\ 0\ 0\ 1]$$

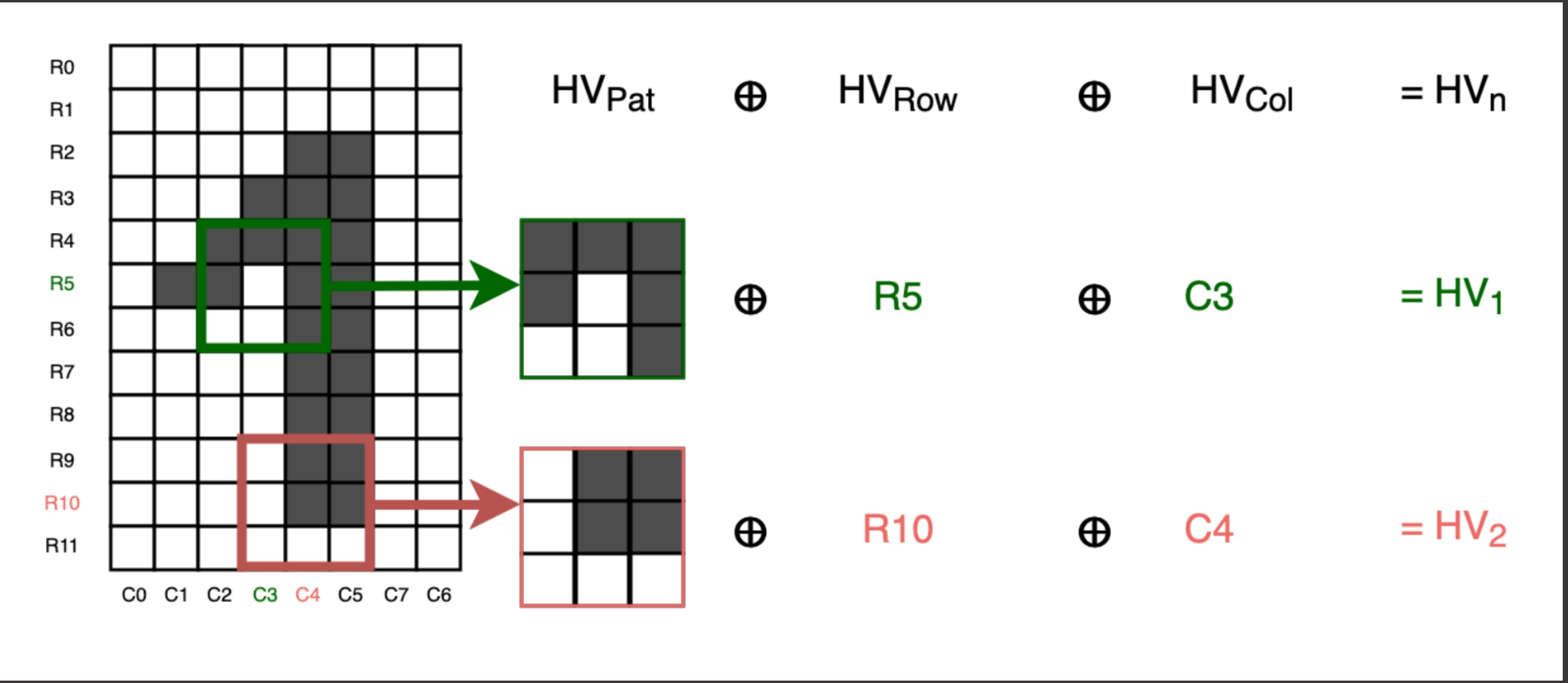
$$C := [0\ 0\ 0\ 0\ 1\ 0\ 1\ 0]$$

$$A \oplus C = A|C = [0\ 0\ 1\ 0\ 1\ 0\ 1\ 1]$$

Bitwise OR

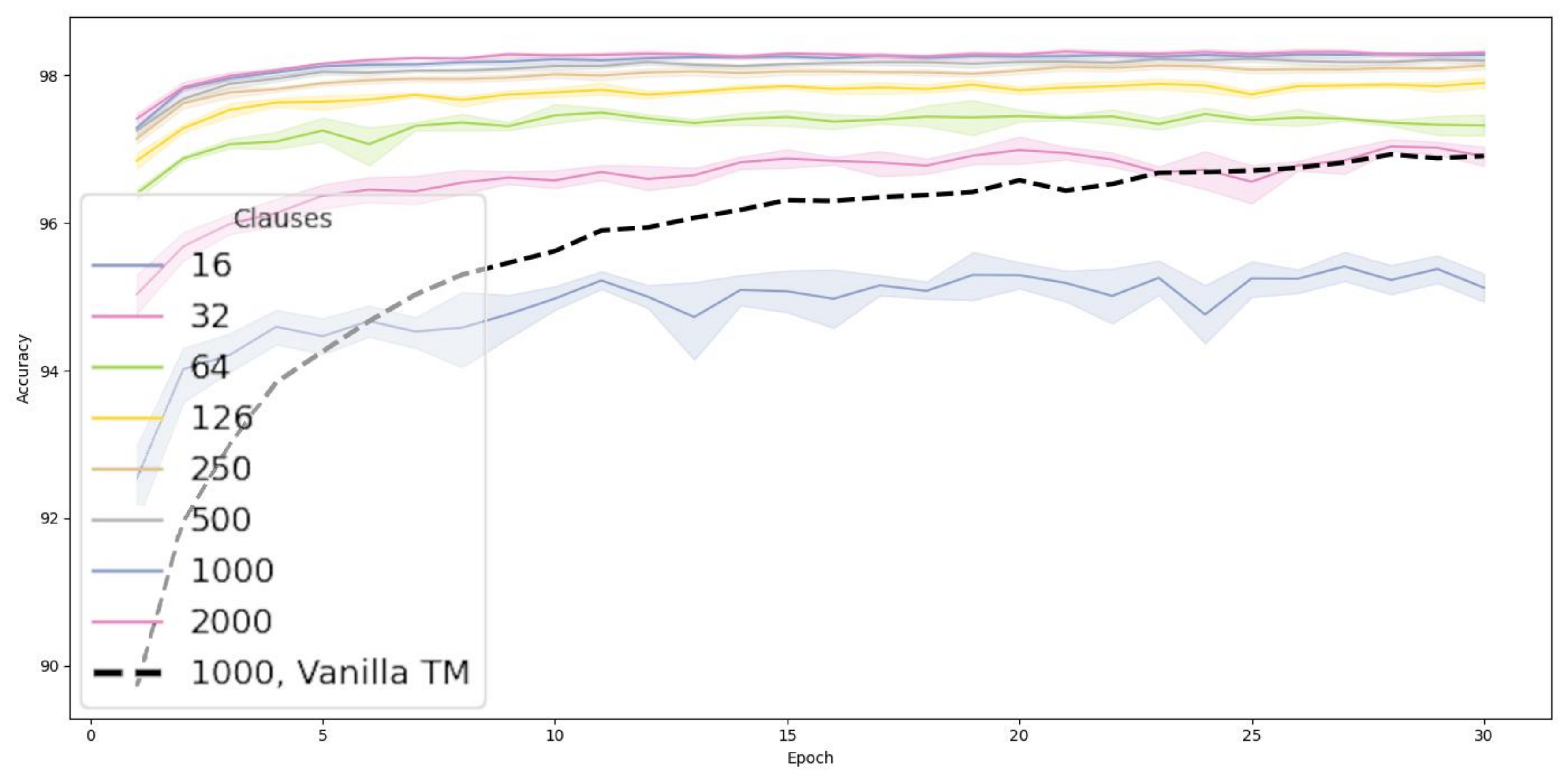
Rachkovskij DA (2001) Representation and processing of structures with binary sparse distributed codes. IEEE Trans Knowl Data Eng 13(2):261–276.

MNIST with 3x3 pixel symbols and row/column bundling



V. Halenka *et al.*, "Exploring Effects of Hyperdimensional Vectors for Tsetlin Machines," *2024 International Symposium on the Tsetlin Machine (ISTM)*, Pittsburgh, US, 2024.

3x3 pixel symbols vs single-pixel symbols



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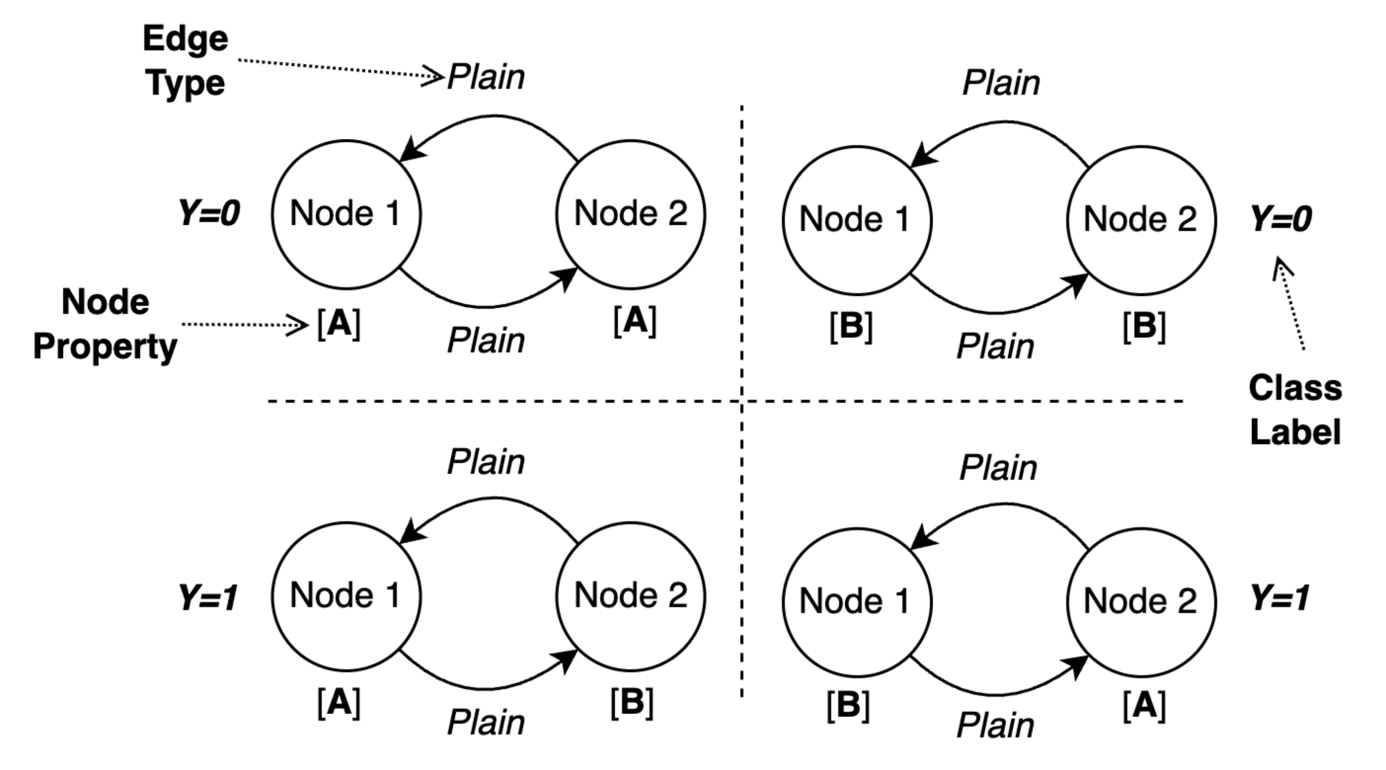
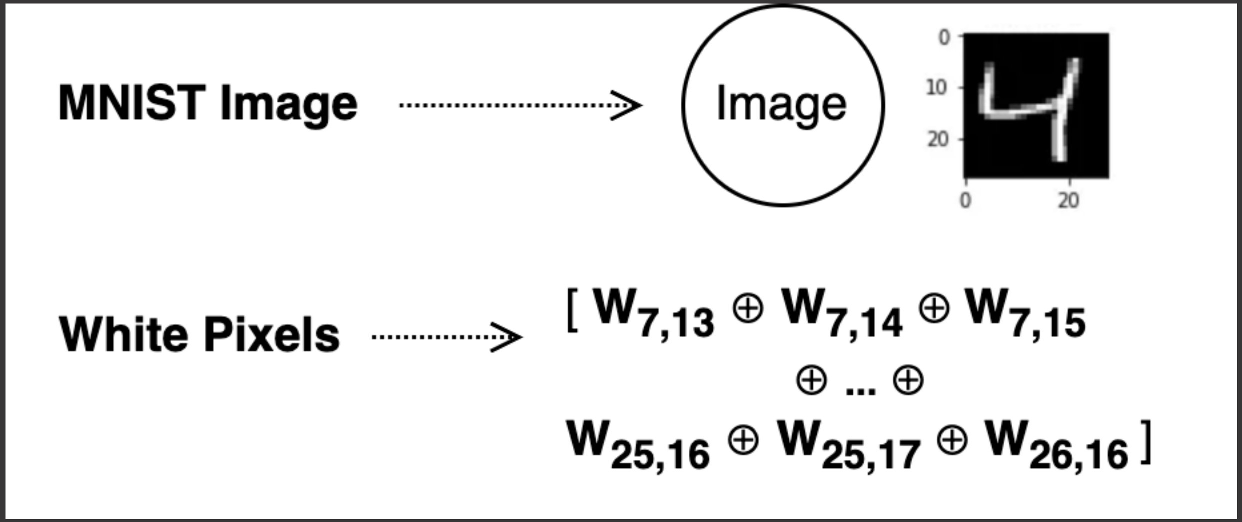
Example problems solved by the Graph Tsetlin Machine

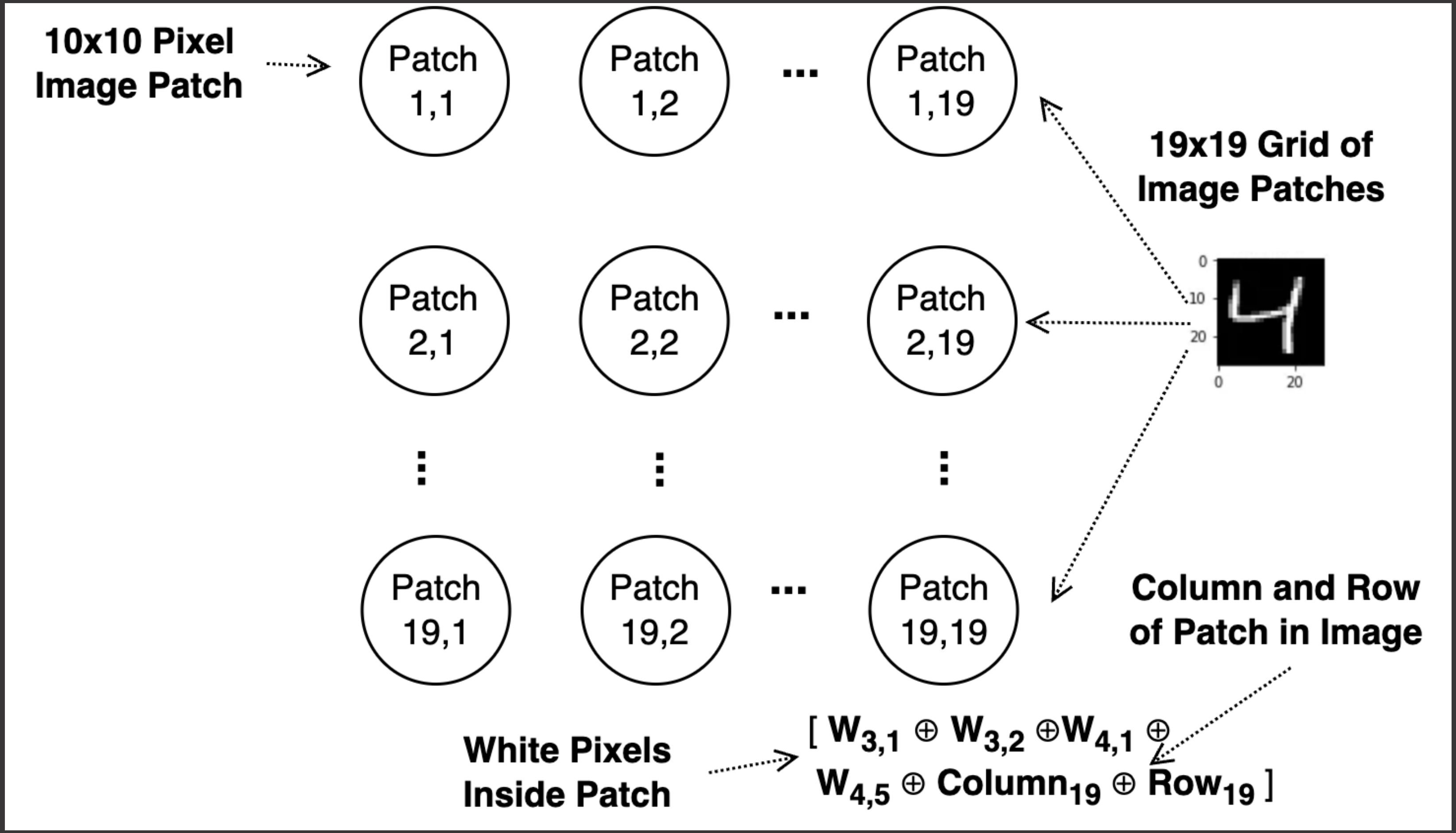
Noisy XOR


Edge Type
Plain
Y=0
Node 1
Node 2
Node Property
[A]
Plain
[A]
Plain
Node 1
Node 2
Y=0
[B]
Plain
[B]
Class Label
Plain
Y=1
Node 1
Node 2
[A]
Plain
[B]
Plain
Node 1
Node 2
Y=1
[B]
Plain
[A]

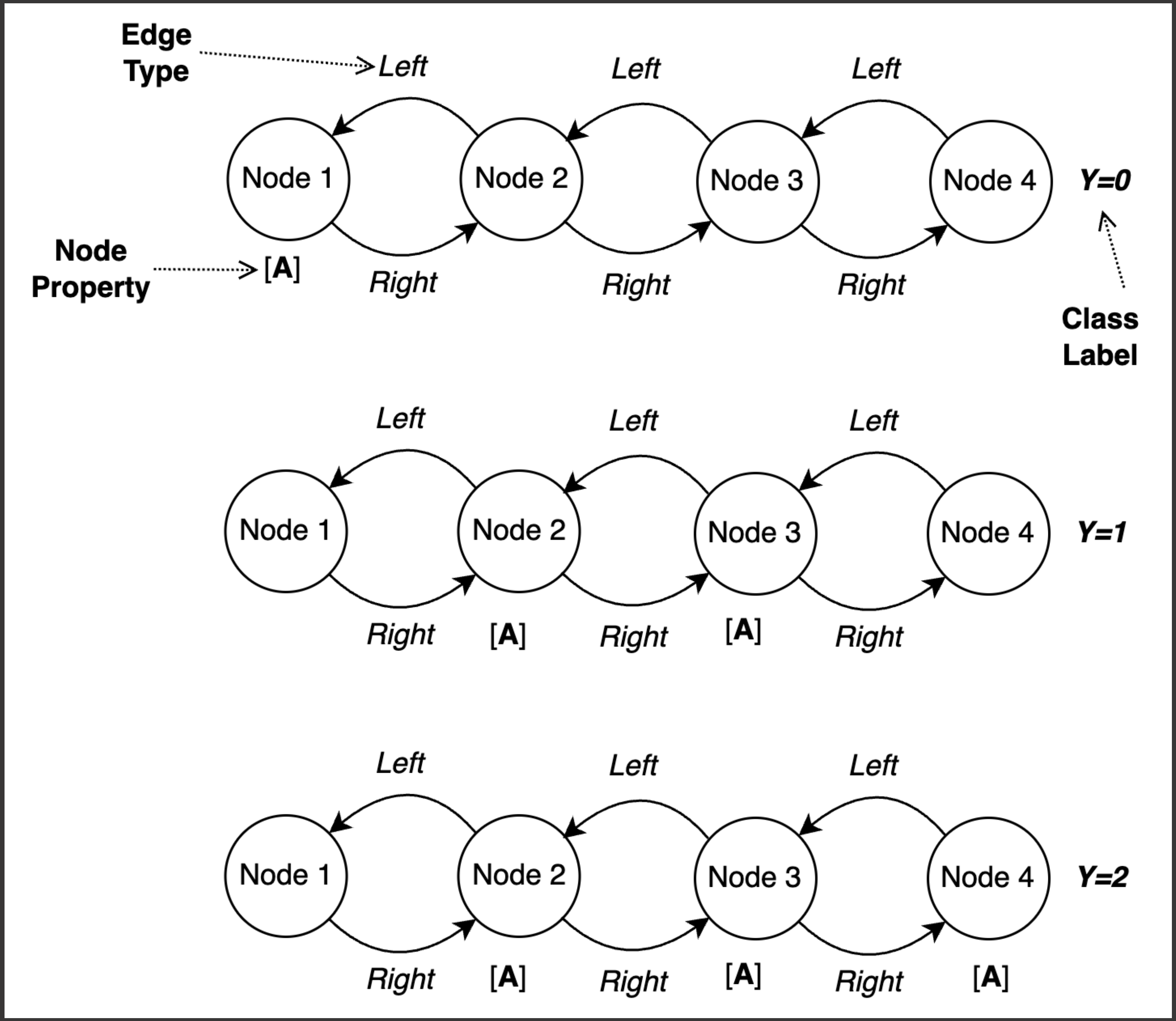
Vanilla MNIST



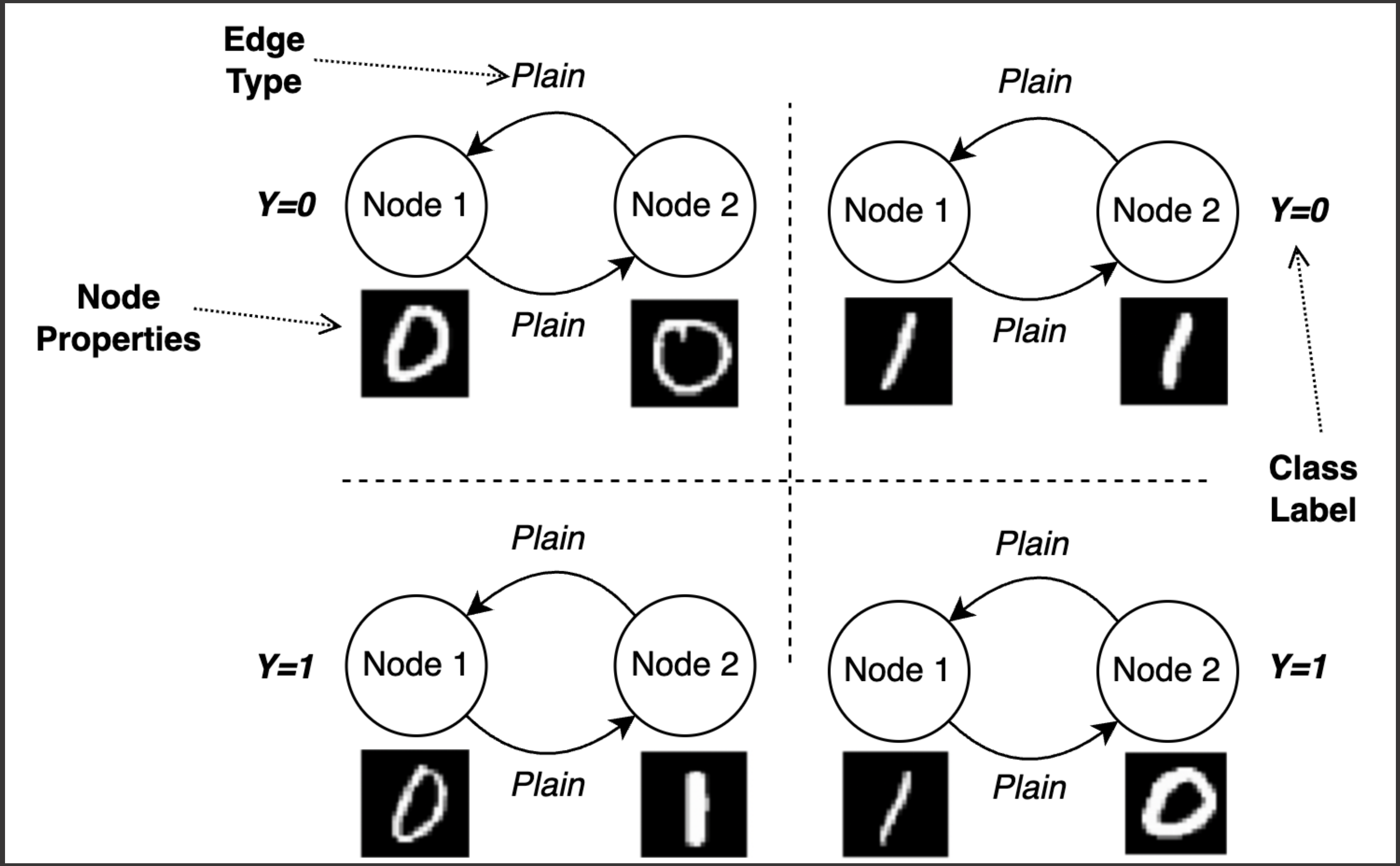
Convolutional MNIST



Sequence classification



MNIST Noisy XOR



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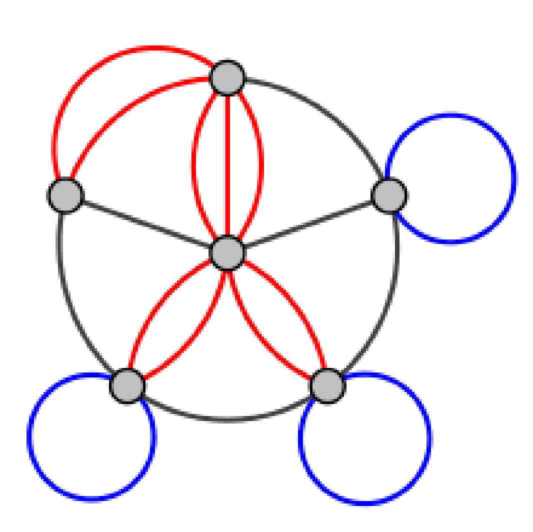
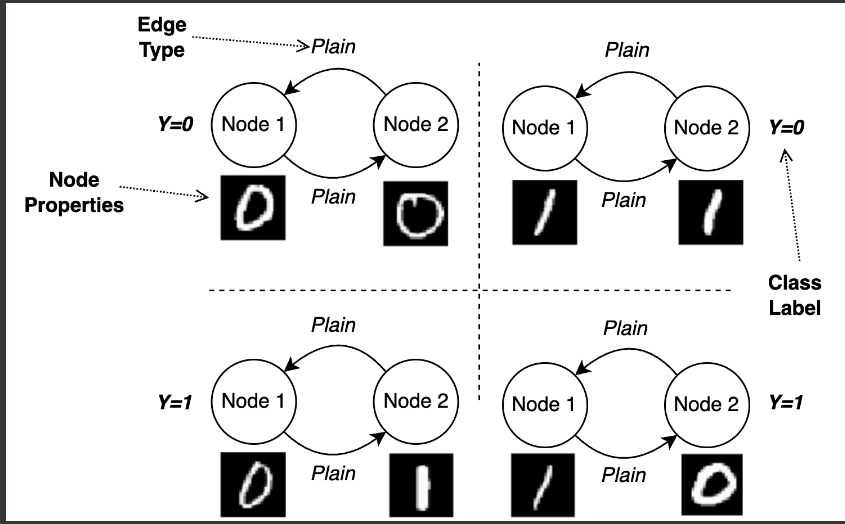
Tsetlin machine for logical learning and reasoning with graphs

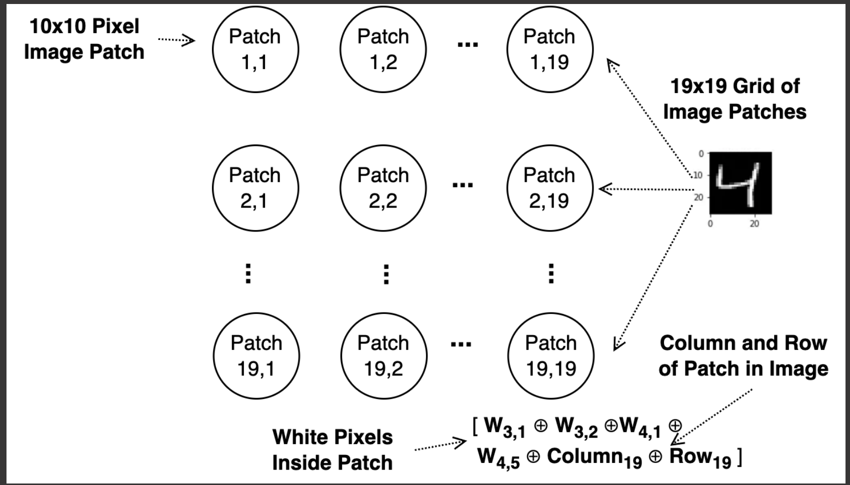
https://github.com/cair/GraphTsetlinMachine



- Processes directed and labeled multigraphs
- Vector symbolic node properties and edge types
- Nested (deep) clauses
- Arbitrarily sized inputs
- Incorporates Vanilla, Multiclass, Convolutional, and Coalesced Tsetlin Machines
- Rewritten faster CUDA kernels



Metric	GraphTM-1	GraphTM-2	BiLSTM	LSTM	GRU	BiLSTM-CNN	GraphCNN
Training Accuracy (%)	60.74	95.17	94.43	88.70	94.68	**96.77**	96.64
Testing Accuracy (%)	59.81	95.14	92.69	87.29	94.05	95.44	**96.35**
Training Time (s)	62.47	84.37	50.39	26.02	**25.47**	32.65	226.36



Classification of viral diseases from nucleotide sequences